How do we decide whether a patch is faster?

Rule for one full measurement vs stock

1. **Measure**
   10 processes, 9 repetitions each
2. **Per process**
   median candidate time / median stock time
3. **Test**
   exact sign-flip test on the 10 ratios
4. **Judge**
   faster only if all three hold

   direction: faster | **p < 0.01** | **reduction ≥ 2%**

Checked against known cases

99/100
Known true time reduction ≈ 4.8%: judged faster

0/500
Synthetic null, no change: false accepts, 95% CP upper bound 0.735%

0/804
Real SDK, no substantive change: false accepts, 95% CP upper bound 0.458%

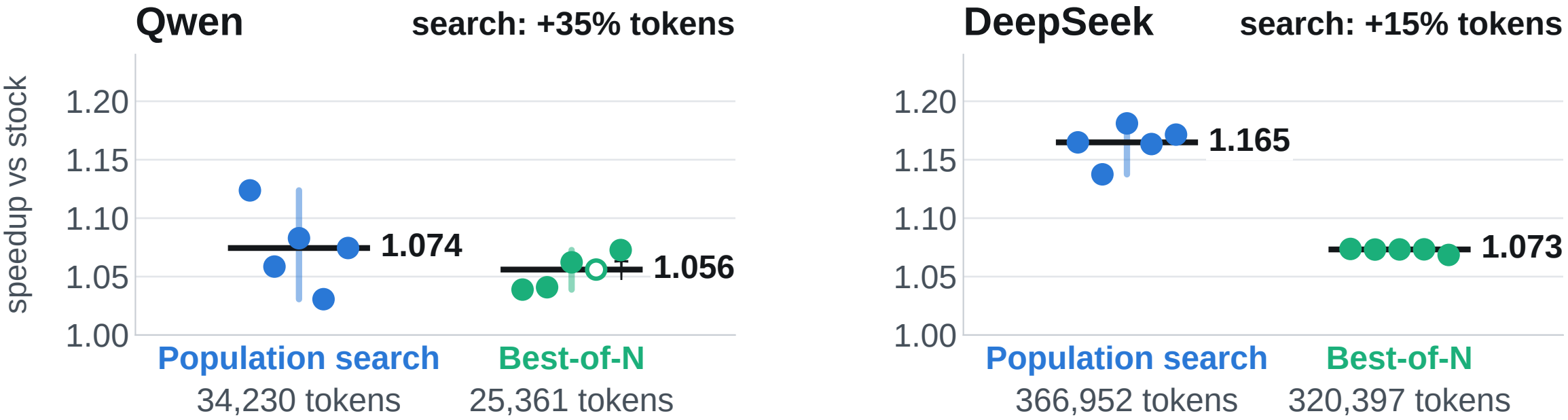
Per single full measurement, in the conditions tested; not the false-positive rate of a whole search. 0 observed does not mean a true rate of zero.

Garmin FIT SDK 21.201.0 | full decode of 4 FIT files | stock and candidates built with -O2

Does search beat independent sampling—and at what cost?

Population (6): pick parent + function → LLM rewrite → compile + checks → timing → keep, feed back failures

C3 matched best-of-N, 40 requests/run C4 5 seeds/configuration dot = median of 3 re-measurements



C1 compiler flags, speedup vs stock

-O2 1.000× -O3 1.090× -O3 -march=native 1.108×

C2 human-guided, AI-assisted patches

1.068× / 1.082×

DeepSeek: all five search results exceeded all five best-of-N results; Qwen showed no clear advantage in this sample.

Requests were matched; token costs were not.

† Qwen best-of-N seed 4: re-measurement passed the rule 2/3. Tokens = reported usage per run (median); requests without usage (DeepSeek HTTP 502) are not counted.
C1/C2: separate single-node batches, 5 measurements each. C2 patches: GetMesg lookup table; GetMesg + GetFieldIndex lookup tables; no formal timing record.

Passing the benchmark did not guarantee equivalent behavior

Supplied checks — before timing every candidate

- 4 fixed FIT files, single thread
- decode checksum + profile digest
- candidate must not start threads

Expanded validation

- 55 extra valid FIT files (written by the SDK encoder)
- 400 mutated inputs
- ASan/UBSan: memory errors, undefined behavior
- TSan: 4 threads, 24 files

Final patches of the 20 runs

20/20 passed the supplied checks

16/20 had findings in the expanded dynamic checks

3/20 differed on additional valid FIT files

12 data-race report only (no wrong output seen)

4 also output difference, crash or memory error

4 no new finding detected (not proof of safety)

Qwen population search, seed 3
Original benchmark files: 4/4 matched
Additional valid files: **45/55 differed**

Broader correctness validation is needed before treating benchmark speedups as reliable optimizations.